

SAVITRIBAI PHULE PUNE UNIVERSITY



A Mini Project Report based on:

Airline scheduling and cargo schedules

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CERTIFICATE

This is to certify that following students from **Third Year Computer Engineering** have successfully completed and submitted the Mini Project work titled **Airline scheduling and cargo schedules** at PDEA's College of Engineering, Manjari (BK), Pune – 412307 in the partial fulfillment of the Bachelor's Degree in Computer Engineering by Savitribai Phule University of Pune for the academic year 2024-2025

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INDEX

Sr. No.	Content	Page No.
1.	Abstract	1
2.	Introduction	2
3.	Objective	3
4.	System Requirement	3
5.	Design and Implementation	4
6.	Code	8
7.	Result/Output	9
8.	Conclusion	10

ABSTRACT

This project aims to develop a dynamic and user-friendly website for a tour and travels company. The website will serve as a platform for users to explore various travel destinations, browse through different tour packages, and make bookings seamlessly. With a focus on enhancing the user experience, the website will feature intuitive navigation, visually appealing design elements, and robust functionality. Additionally, it will incorporate features such as online booking forms, interactive maps, and informative content about destinations and activities. This paper presents a simplified Expert System for airline scheduling, designed to optimize flight operations by managing aircraft availability, crew assignments, and airport constraints. The system comprises a knowledge base that includes data on aircraft status, crew schedules, and flight details. An inference engine applies rule-based logic to suggest optimal schedules, ensuring efficient resource utilization. The implementation features a basic Python framework that facilitates the addition of aircraft, crew, and flights, while scheduling logic evaluates availability and constraints. This system serves as a foundational tool for enhancing airline operational efficiency, with potential for further development in complexity and user interaction.

INTRODUCTION

In the vast expanse of the digital world, where billions of users connect, communicate, and collaborate every day, web technology stands as the cornerstone of this interconnected realm. At its core, web technology encompasses an intricate web of tools, languages, and protocols that empower individuals and businesses to create, share, and consume information seamlessly across the internet. In the rapidly evolving aviation industry, effective scheduling is critical for maximizing efficiency and minimizing operational costs. Airline scheduling involves coordinating flights, aircraft, and crew members to meet passenger demand while adhering to regulatory constraints and resource availability. This complexity necessitates the implementation of advanced systems that can automate and optimize scheduling processes.

This paper introduces a simplified Expert System designed specifically for airline scheduling. The system leverages a structured knowledge base to manage crucial data related to aircraft, crew, and flight schedules. By employing a rule-based inference engine, the system evaluates various constraints and generates optimal flight schedules. The goal is to enhance operational efficiency and ensure timely service, ultimately improving the overall travel experience for passengers. As the aviation sector continues to grow, such systems will play an increasingly vital role in streamlining operations and supporting strategic decision-making.

OBJECTIVE

1. The primary objective of this Expert System for airline scheduling is to optimize the allocation of resources, including aircraft and crew, to ensure efficient and timely flight operations. By automating the scheduling process, the system aims to:
2. Enhance Operational Efficiency:
Streamline the scheduling workflow to minimize delays and maximize aircraft utilization.
3. Improve Resource Management:
Effectively manage aircraft availability and crew assignments while adhering to regulatory constraints and duty limits.
4. Increase Flexibility:
Allow for quick adjustments to schedules in response to changing conditions, such as weather disruptions or last-minute flight requests.
5. Support Decision-Making:
Provide actionable insights and recommendations to airline operators, facilitating informed decision-making and strategic planning.
6. Enhance Customer Satisfaction:
Ensure timely and reliable flight schedules, thereby improving the overall passenger experience

SYSTEM REQUIREMENTS:

1. **Hardware Requirements:** Servers, Mobile Devices
2. **Software Requirements:** ERP, FMS
3. **Technology Stack:** Java, Python, HTML, CSS, AWS

DESIGN AND IMPLEMENTATION

System Architecture:

The Expert System for airline scheduling is designed with a modular architecture that separates the knowledge base, inference engine, and user interface:

- **Knowledge Base:** Stores data about aircraft, crew, and flight schedules. It uses structured data formats (e.g., JSON or databases) for easy retrieval and management.
- **Inference Engine:** Processes the rules and logic for scheduling. It evaluates constraints, such as aircraft availability and crew duty limits, and generates optimal flight schedules.
- **User Interface:** Provides a web-based or desktop interface for operators to input data, view schedules, and receive recommendations.

Introduction:

Airline scheduling and cargo logistics are complex operations that require the coordination of multiple interdependent variables, such as aircraft availability, crew working hours, airport slots, weather conditions, and cargo capacity. With the growing demand for timely and efficient air transport, airlines are leveraging intelligent expert systems to automate and optimize their scheduling processes. This document outlines the design and implementation of an expert system for airline scheduling and cargo schedules, highlighting the modular architecture and practical workflow of the system.

System Overview:

The expert system is structured around three core components: the Knowledge Base, the Inference Engine, and the User Interface. Each component plays a vital role in ensuring accurate scheduling and efficient cargo management.

- **Knowledge Base:** Acts as the system's memory, storing structured information on aircraft types, crew schedules, airport data, maintenance windows, and cargo constraints. This component ensures that all necessary data is readily available for decision-making.

- **Inference Engine:** The brain of the system, it applies rules and logic to the data from the knowledge base. It considers real-world constraints such as crew rest periods, aircraft turnaround times, airport slot availability, and cargo capacity limits to generate optimal flight and cargo schedules.
- **User Interface:** A user-friendly web-based or desktop application that enables airline operators to interact with the system. It allows for data input, visualization of schedules, conflict alerts, and recommendations for rescheduling.

The system follows a modular architecture to promote flexibility, scalability, and ease of maintenance. Here is a breakdown of the modules:

- **Data Management Module:** Handles the retrieval, storage, and update of scheduling data. Integrates with external data sources like weather APIs and flight tracking systems.
- **Rule Processing Module:** Hosts the rules and constraints used by the inference engine. Uses logic programming or a rules engine to evaluate and apply scheduling policies.
- **Optimization Module:** Uses algorithms such as linear programming, genetic algorithms, or heuristic methods to find the most efficient scheduling combinations.
- **Visualization Module:** Presents schedules, conflicts, and cargo allocations using charts, timelines, and interactive dashboards.

Implementation Details

- **Technologies Used:**
 - Backend: Python with Flask or Django
 - Frontend: React.js or Vue.js for dynamic interfaces
 - Database: MySQL or MongoDB for storing structured data
 - Optimization Libraries: PuLP, Google OR-Tools
 - Deployment: Docker for containerization and deployment on cloud or local servers
- **Workflow:**
 1. Operators input flight, crew, and cargo data.
 2. The system processes data using the inference engine.
 3. Constraints and rules are evaluated.

4. An optimized schedule is generated and displayed.
5. The operator reviews and approves the schedule.

Cargo Scheduling Integration

Cargo scheduling is seamlessly integrated with the flight scheduling system. It includes:

- Real-time weight and volume tracking
- Cargo priority levels (e.g., perishable, urgent, standard)
- Compatibility checks with aircraft type
- Automatic allocation of cargo to suitable flights
- Delay and re-routing management in case of disruptions

Advantages of the System

- Reduces human error in scheduling
- Increases operational efficiency
- Enables quick rescheduling in case of disruptions
- Ensures regulatory compliance (e.g., crew duty limits)
- Provides real-time insights and decision support

This expert system for airline and cargo scheduling demonstrates how AI-driven logic and modular software architecture can address real-world aviation logistics challenges. With the growing complexity of air travel and freight operations, such systems are becoming essential for achieving optimal performance, reducing costs, and enhancing reliability. Future enhancements may include integration with AI/ML models for predictive scheduling and the use of IoT devices for real-time cargo tracking and aircraft monitoring.

SOURCE CODE

```
class AirlineManagementSystem: def
    __init__(self): self.flights = {}
    self.cargo = {} self.passengers =
        {}

    def add_flight(self, flight_id, origin, destination, date, time): self.flights[flight_id] = {'origin': origin,
        'destination': destination, 'date': date,
        'time': time} print(f"Flight added: {flight_id} from {origin} to {destination} on {date} at {time}.")

    def view_flights(self):
        if not self.flights:
            print("No flights scheduled.") return
        for flight_id, details in self.flights.items(): print(f"Flight {flight_id}: {details['origin']} ->
        {details['destination']} on {details['date']} at {details['time']}")

    def add_cargo(self, cargo_id, description, weight): self.cargo[cargo_id] = {'description':
        description, 'weight': weight} print(f"Cargo added: {cargo_id} - {description}, Weight:
        {weight} kg")

    def track_cargo(self, cargo_id):
        return self.cargo.get(cargo_id, "Cargo not found.")

    def book_passenger(self, passenger_id, name, flight_id):
        if flight_id in self.flights:
            self.passengers[passenger_id] = {'name': name, 'flight_id': flight_id} print(f"Passenger booked:
            {name} for flight {flight_id}.") else:
            print("Flight not found.")

# Example usage if __name__ == "__
main_":
```

```
airline_system = AirlineManagementSystem()

# Adding flights
airline_system.add_flight("AA101", "New York",
    "London", "2025-05-01",
    "14:00")
airline_system.add_flight("BA202", "London", "Tokyo", "2025-05-02", "10:00")

# Viewing flights
airline_system.view_flights()

# Adding cargo
airline_system.add_cargo("CARGO001", "Electronics", 500)
airline_system.add_cargo("CARGO002", "Clothing", 200)

# Tracking cargo
print(airline_system.track_cargo("CARGO001"))
print(airline_system.track_cargo("CARGO003"))

# Booking passengers
airline_system.book_passenger("P001", "Alice Johnson", "AA101")
airline_system.book_passenger("P002",
    "Bob Smith", "BA202")
```

RESULT

```
Flight added: AA101 from New York to London on 2025-05-01 at 14:00.  
Flight added: BA202 from London to Tokyo on 2025-05-02 at 10:00.  
Flight AA101: New York -> London on 2025-05-01 at 14:00  
Flight BA202: London -> Tokyo on 2025-05-02 at 10:00  
Cargo added: CARGO001 - Electronics, Weight: 500 kg  
Cargo added: CARGO002 - Clothing, Weight: 200 kg  
{'description': 'Electronics', 'weight': 500}  
Cargo not found.  
Passenger booked: Alice Johnson for flight AA101.  
Passenger booked: Bob Smith for flight BA202.
```

CONCLUSION

The development of the Expert System for airline scheduling represents a significant advancement in optimizing flight operations within the aviation industry. By leveraging a structured knowledge base and a rule-based inference engine, the system effectively manages the complexities of scheduling, including aircraft availability, crew assignments, and regulatory constraints.